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# The Omitted Dimension: Exploiting Multiuser Diversity in Multi-Radio Access Technology Data Cellular Communication Systems

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**ABSTRACT** Performance of cellular communication systems is typically enhanced by leveraging the three dimensions of system transmission bandwidth, area frequency reuse, i.e. system access point (AP) density, and spectral efficiency of employed radio air interface, referred to as the radio access technology (RAT). In particular, refined RATs are continuously introduced to compensate for limited spectrum availability and restrictions on system AP densification in cellular communication systems. While the performance of cellular communication systems is maximized by employing a single RAT providing maximal spectral efficiency, varying capabilities of user equipment compels the co-deployment of multiple RATs in cellular communication systems and entails fragmenting system radiofrequency resources between co-deployed RATs. Nevertheless, the inefficient structuring of multi-RAT systems as independently operated collocated single-RAT subsystems results in unbalanced system loading, suboptimal spectrum utilization, and the omission of multiuser diversity as a performance enhancement dimension in multi-RAT systems. The omitted dimension of multiuser diversity is exploited in this paper, through multiple means and techniques, to further enhance the performance of multi-RAT data cellular communication systems. Unifying the architectural structure of multi-RAT systems is proposed to eliminate the redundant duplication of radio access network functions and elements, reduce system deployment costs and operational complexity, improve system scalability, and enable the joint execution of non-radio transmission functions for all co-deployed RATs. By fully exploiting system multiuser diversity, the joint allocation of system radiofrequency resources under autonomous spectrum assignment is shown to substantially enhance the performance of all employed RATs, in addition to the overall performance of multi-RAT systems, without extending any of the three typical performance enhancement dimensions of cellular communication systems.

**INDEX TERMS** Cellular communication systems, multi-radio access technology systems, radio access networks, radio access technologies, radio resource allocation, spectrum management, user access, user equipment.

## I. INTRODUCTION

Mobile network operators deploy the Network Infrastructure (NI) of cellular communication systems, upon obtaining a spectrum utilization license, to wirelessly connect User Equipment (UE) employed by mobile system users [1], [2]. As Fig. 1 shows, the NI of a cellular communication system is generally divided into a Radio Access Network (RAN) and a Core Network (CN) [2]–[6]. RANs wirelessly cover specific geographic areas by deploying fixed Access Points (APs) and facilitate UE access to system radiofrequency resources using a wireless radio interface referred to as the Radio Access Technology (RAT) [6]–[9], with the terms 'radiofrequency resources' and 'spectrum' interchangeably used to denote the aggregate frequency bands utilized by a cellular system. At the other end of the RAN, backhaul links transport user traffic to the CN, primarily comprising data routers and core links interconnecting cellular systems with other communication systems [2]–[6].

RATs are characterized by the transmission bandwidth, transmission frame duration, frequency reuse factor between

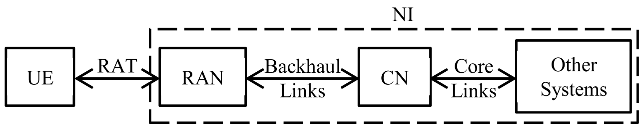


**FIGURE 1.** Architectural structure of cellular communication systems.

system APs, user multiple access scheme, modulation and coding configurations along with the signaling and transmission protocols [7]–[9]. To exploit the significant variations and fluctuations in achievable user throughput rates, data cellular communication systems typically utilize opportunistic scheduling techniques for allocation of system radiofrequency resources [10]–[21]. Specifically, Proportional Fair Scheduling (PFS) [22] is employed in data cellular systems to balance the compromise between maximizing the overall system performance and attaining fairness between system users [10]–[22]. Such balance is achieved by allocating system radiofrequency resources to users maximizing a time-varying utility function of the ratio between achievable and averaged user throughput rates, with the system performance improving as the number of users, and thus the diversity in user performance, increases [10]–[22].

The performance of cellular communication systems is typically dimensioned by the system transmission bandwidth, area frequency reuse, i.e. system AP density, and spectral efficiency of employed RAT [1], [2]. With limited spectrum availability and restrictions on system AP densification, refined RATs with higher spectral efficiency are continuously introduced to enhance the performance of cellular communication systems [23]–[25]. Maximizing the performance of a cellular communication system thus requires employing the most spectrally efficient RAT to connect all system users. However, large variations in UE capabilities compels the co-deployment of multiple RATs to maintain user connectivity in cellular communication systems [23]–[25]. Nevertheless, the design and structuring of classical cellular communication systems assumes the employment of a single RAT to connect system users [1]–[3]. Such an assumption implies fixed user diversity and effectively eliminates the feasibility of exploiting system multiuser diversity as a performance enhancement dimension in multi-RAT cellular communication systems. On the other hand, structuring multi-RAT systems as groups of independently operated collocated single-RAT subsystems leads to a multitude of inefficiencies in system operation, spectrum management and user access [23]–[31].

Building on the concept introduced in [29], the aim of this paper is to address the shortcomings subsequent to the disjoint structuring and independent operation of co-deployed RATs in multi-RAT systems by providing a unified framework for system structuring and operation, user access, spectrum management and radiofrequency resource allocation. The rest of the paper is organized as follows. Section II discusses the shortcomings of the classical structural modelling for multi-RAT systems while section III provides refinements to the design principles of multi-RAT systems that maintain independent RAT operation. Section IV details the unified structuring and RAT operation framework for multi-RAT systems and the simulation setup for performance evaluation is specified in Section V, with Section VI comparing the performance of different system structuring configurations. Section VII Concludes the paper. The acronyms and symbols used in this paper are detailed in Tables 1 and 2, respectively.

**TABLE 1.** List of acronyms.

| | |
|---|---|
| 3GPP | 3GPP: 3rd Generation Partnership Project |
| AP | Access Point |
| ASA | Autonomous Spectrum Assignment |
| AWGN | Additive White Gaussian Noise |
| CN | Core Network |
| FSA | Fixed Spectrum Assignment |
| HSPA | High Speed Packet Access |
| IRA | Independent Resource Allocation |
| ITU | International Telecommunication Union |
| JRA | Joint Resource Allocation |
| LTE | Long Term Evolution |
| MCS | Modulation Coding Scheme |
| MMA | Multimode Access |
| MMUE | Multimode User Equipment |
| NI | Network Infrastructure |
| PFS | Proportional Fair Scheduling |
| RAN | Radio Access Network |
| RAT | Radio Access Technology |
| RTF | Radio Transmission Function |
| SINR | Signal to Interference plus Noise Ratio |
| SMA | Single-Mode Access |
| SMUE | Single-Mode User Equipment |
| UE | User Equipment |

## II. MULTI-RADIO ACCESS TECHNOLOGY CELLULAR COMMUNICATION SYSTEMS

### A. RADIO ACCESS NETWORK STRUCTURING AND OPERATION

The typical RAN structuring of multi-RAT systems under independent RAT operation, referred to as the disjoint RAN structuring and illustrated in Fig. 2, is broken down into independent collocated single-RAT sub-RANs. RAN functions, broadly divided into Radio Transmission Functions (RTFs) and non-RTFs as detailed in Table 3, are independently performed for each RAT under disjoint RAN structuring. In particular, system radiofrequency resources are independently allocated for each of the employed RATs under Independent Resource Allocation (IRA). The deployment time, costs and operational complexity of multi-RAT systems adopting disjoint RAN structuring thus scales with the number of employed RATs. Furthermore, multi-RAT systems adopting disjoint RAN structuring suffer from limited scalability; as RAN functionality is independently enhanced for each employed RAT while the introduction of a new RAT is equivalent to a new RAN deployment. In addition, the introduction and integration of joint multi–RAT functions is also complicated by disjoint RAN structuring [23], [24], [27].

**TABLE 2.** List of symbols.

| | | | |
|---|---|---|---|
| $\alpha$ | Transmission attenuation factor | $K_T$ | Total number of system users |
| $\alpha_i$ | Transmission attenuation factor for RAT $i$ | $K_{T,i}$ | Total number of system users utilizing RAT $i$ as the primary mode of operation |
| $\beta$ | Maximal spectral efficiency | $L_F$ | Log-normal shadowing loss |
| $\theta$ | Azimuth angle $j$ | $L_P$ | Path loss |
| $\theta_{3dB}$ | AP antenna 3 dB beamwidth | $M_F$ | Channel Gain |
| $\xi$ | SINR | $N_F$ | Noise Figure |
| $\xi_{\max}$ | SINR Required to utilize highest performing MCS | $P_t$ | Transmitted power |
| $\xi_{\min}$ | SINR Required to utilize lowest performing MCS | $P_r$ | Received power |
| $A_{\max}$ | Maximum AP antenna attenuation | $R$ | Overall average cell throughput rate under FSA, SMA, IRA |
| $B$ | Number of pooled spectrum blocks under ASA | $R_i$ | Average cell throughput rate for RAT $i$ under FSA, SMA, IRA |
| $B_i$ | Number of pooled spectrum blocks assigned to RAT $i$ under ASA | $R_k$ | Overall average throughput rate of user $k$ under FSA, SMA, IRA |
| $C_i$ | Carrier transmission bandwidth for RAT $i$ | $R_{k,i}$ | Average throughput rate of user $k$ utilizing RAT $i$ as the primary mode of operation under FSA, SMA and IRA |
| $D_i$ | Bandwidth continuously assigned to RAT $i$ under ASA | $R\|_{ASA}$ | Overall average cell throughput rate under ASA |
| $d$ | Distance | $R_i\|_{ASA}$ | Average cell throughput rate for RAT $i$ under ASA |
| $G_i\|_{ASA}$ | Gain in the average cell throughput rate of RAT $i$ due to the employment of ASA over FSA | $R_k\|_{ASA}$ | Overall average throughput rate of user $k$ under ASA |
| $G_k\|_{ASA}$ | Gain in the overall average user throughput rate due to the employment of ASA over FSA | $R_{k,i}\|_{ASA}$ | Average throughput rate of user $k$ utilizing RAT $i$ as the primary mode of operation under ASA |
| $G_T\|_{ASA}$ | Gain in the overall average cell throughput rate due to the employment of ASA over FSA | $R\|_{JRA}$ | Overall average cell throughput rate under JRA |
| $G_i\|_{JRA}$ | Gain in the average cell throughput rate of RAT $i$ due to the employment of JRA over IRA | $R_i\|_{JRA}$ | Average cell throughput rate for RAT $i$ under JRA |
| $G_k\|_{JRA}$ | Gain in the overall average user throughput rate due to the employment of JRA over IRA | $R_k\|_{JRA}$ | Overall average throughput rate of user $k$ under JRA |
| $G_T\|_{JRA}$ | Gain in the overall average cell throughput rate due to the employment of JRA over IRA | $R_{k,i}\|_{JRA}$ | Average throughput rate of user $k$ utilizing RAT $i$ as the primary mode of operation under JRA |
| $G_i\|_{MMA}$ | Gain in the average cell throughput rate of RAT $i$ due to the employment of MMA over SMA | $R\|_{MMA}$ | Overall average cell throughput rate under MMA |
| $G_k\|_{MMA}$ | Gain in the overall average user throughput rate due to the employment of MMA over SMA | $R_i\|_{MMA}$ | Average cell throughput rate for RAT $i$ under MMA |
| $G_T\|_{MMA}$ | Gain in the overall average cell throughput rate due to the employment of MMA over SMA | $R_k\|_{MMA}$ | Overall average throughput rate of user $k$ under MMA |
| $g$ | AP antenna gain | $T_i$ | Transmission frame duration for RAT $i$ |
| $g_{\max}$ | Maximum AP antenna gain | $TA_{min}$ | Minimal pooled spectrum block assignment interval under ASA |
| $H_K$ | Multiuser diversity factor of order $K$ | $W$ | Total Transmission Bandwidth |
| $I$ | Number of Co-deployed RATs | $W'$ | ASA Transmission budget |
| $\boldsymbol{I_k}$ | Set of RATs jointly supported by the connecting AP and the UE transceiver module of user $k$ | $W_B$ | Pooled spectrum block bandwidth under ASA |
| $K$ | Number of users at the considered system cell | $W_i$ | Transmission bandwidth assigned to RAT $i$ at the considered cell under FSA |
| $K_i$ | Number of users utilizing RAT $i$ as the primary mode of operation at the considered cell | $W_i^*$ | Transmission bandwidth assigned to RAT $i$ at the considered cell under ASA |
| $K'_i$ | Number of users capable of utilizing RAT $i$ at the considered cell | | |

### B. USER ACCESS TO SYSTEM RADIOFREQUENCY RESOURCES

Multiple RATs are primarily co-deployed in cellular communication systems to maintain connectivity of UE capable of utilizing a single RAT only, referred to as Single-Mode UE (SMUE). On the other hand, Multimode UE (MMUE) are equipped with multimode transceiver modules, capable of utilizing multiple RATs, to maintain user connectivity when the highest performing RAT supported by MMUE is not employed by the connecting AP [26], [28]. While hardware capabilities limit SMUE to utilizing a single RAT when connecting to system APs, connectivity of MMUE to system APs is also limited to a single RAT only by choice of user access policy under Single-Mode Access (SMA). Nevertheless, limiting MMUE to utilizing a single RAT only when connecting to system APs results in the suboptimal utilization of system radiofrequency resources and MMUE capabilities [28]. When constrained by SMA, MMUE utilize the highest performing RAT jointly supported by the connecting AP and the MMUE transceiver module, referred to as the primary mode of operation and illustrated in Fig. 3.

### C. SPECTRUM MANAGEMENT

IRA entails apportioning system radiofrequency resources between co-deployed RATs, with system-level Fixed Spectrum Assignment (FSA) typically employed in multi-RAT systems [23], [27]. Under system-level FSA, a fixed transmission bandwidth is assigned to each of the employed RATs

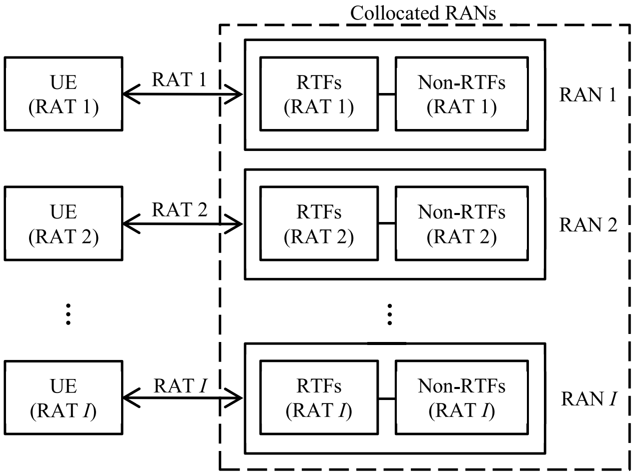


**FIGURE 2.** Disjoint RAN structuring for a multi-RAT system employing *I* RATs.

**TABLE 3.** Main radio access network functions [4], [29].

| | | |
|---|---|---|
| **Radio Transmission Functions (RTFs)** | | - Wireless signal transmission and reception<br>- Modulation and demodulation of carrier waveforms<br>- Baseband signal processing<br>- UE/AP synchronization in frequency and time |
| Non-RTFs | System Access Control (SAC) | - System information broadcast<br>- Admission of authenticated system users |
| | Radio Resource Management (RRM) | - Facilitation of user access to system radio resources |
| | User Mobility Management (UMM) | - User connection transfer between different radio channels and system APs<br>- Paging system users to contact the RAN<br>- System user location positioning |

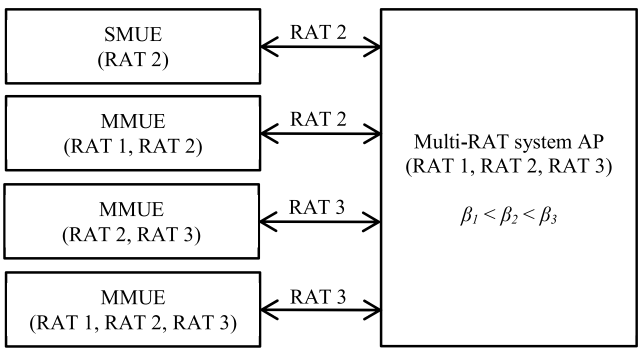


**FIGURE 3.** An example of SMA in multi-RAT systems. In this example, a deployment scenario in which three RATs are co-deployed to connect a SMUE and three MMUE is illustrated. RAT performance is assumed to increase monotonically, and, due to the restriction of SMA, MMUE utilize the primary mode of operation only when connecting to the system AP.

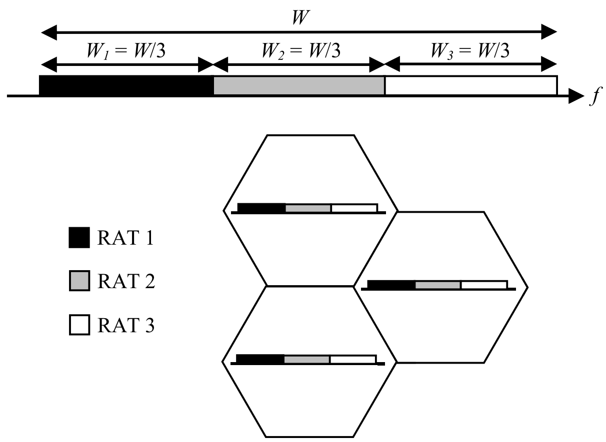


**FIGURE 4.** An example of spectrum assignment under system-level FSA in multi-RAT systems. In this example, a deployment scenario in which three RATs are co-deployed at three system APs is illustrated, and system radiofrequency resources equally divided between employed RATs. Fixed spectrum partitioning between co-deployed RATs is applied at all system APs under system-level FSA.

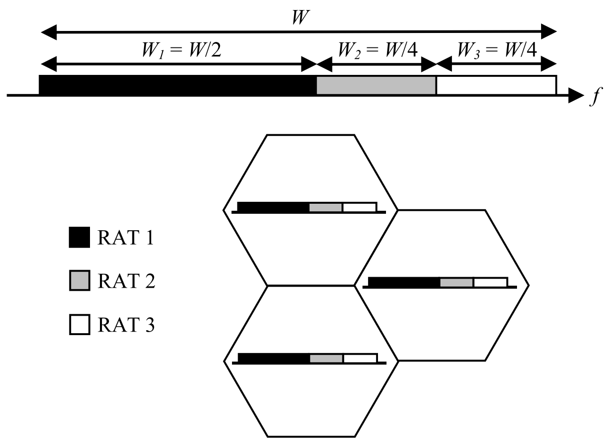


**FIGURE 5.** An example of spectrum refarming in multi-RAT systems, in which the same setup adopted by Fig. 4 is assumed. When compared to Fig. 4, spectrum partitioning between co-deployed RATs is modified to provide RAT 1 with additional radiofrequency resources at the expense of reducing the amount of spectrum originally assigned to RAT 2 and RAT 3. As in system-level FSA, spectrum partitioning between co-deployed RATs is applied at all system APs.

such that all system APs have the same spectrum partitioning between co-deployed RATs as Fig. 4 shows. However, multi–RAT systems are subject to rapid traffic fluctuations and variations between system APs as system users relocate and upgrade UE to take advantage of higher performing RATs [23], [24]. System-level FSA thus results in unbalanced system loading and suboptimal spectrum utilization in multi–RAT systems; as certain RATs may experience excessive traffic demand at certain APs while other RATs are underutilized and vice versa [23], [24], [27], [29], [30]. To align spectrum assignment with changing traffic conditions in multi–RAT systems, spectrum assigned to RATs experiencing long-term declining traffic demand is deducted and reallocated at the system level to RATs with continuously increasing traffic demand, a process illustrated in Fig. 5 and referred to as spectrum refarming [23]. Efforts for enhancing spectrum management in multi–RAT systems are primarily centered at the development of automated and adaptive spectrum refarming techniques [23], [32]–[37]. However, having a small number of system APs with high traffic demand for a particular RAT limits the amount of spectrum that can be refarmed from such a RAT under system-level spectrum assignment. Therefore, the effectiveness of adaptive spectrum refarming techniques is severely reduced by system-level spectrum assignment [23], [27].

### D. SYSTEM PERFORMANCE

The performance of a system cell in a multi-RAT system co-deploying $I$ RATs to connect $K$ users generating infinitely-backlogged, delay-tolerant traffic is considered. Let $K_i$ denote the number of users utilizing RAT $i$ as the primary mode of operation, with $\Sigma_I K_i = K$. Under FSA, a fixed transmission bandwidth, $W_i$, is assigned to RAT $i$ such that $\Sigma_I W_i = W$, with $W_i$ chosen to reflect the traffic demand for RAT $i$. Assuming equal user traffic demand and system-level spectrum assignment, $W_i$ is set to the value

$$W_i = \frac{K_{T,i}}{K_T} W. \tag{1}$$

in which $K_{T,i}$ denotes the total number of system users utilizing RAT $i$ as the primary mode of operation, with $\Sigma_I K_{Ti} = K_T$. Under PFS, the average cell throughput rate for RAT $i$, $R_i$, assuming system-level FSA, SMA and IRA is quantified as [17]–[21]

$$R_i = \alpha_i W_i H_{K_i}. \tag{2}$$

in which $\alpha_i$ denotes the transmission attenuation factor of RAT $i$ and $H_{K_i}$ denotes the multiuser diversity factor of order $K_i$. Precise characterization of $H_{K_i}$ is dependent on the exact system layout, employed RAT and radio resource allocation policies, wireless propagation conditions and user distribution [10]–[21]. The average throughput rate for user $k$ utilizing RAT $i$ as the primary mode of operation, $R_{k,i}$, is thus equal to

$$R_{k,i} = \frac{R_i}{K_i} = \alpha_i W_i \frac{H_{K_i}}{K_i}. \tag{3}$$

Therefore, the overall average cell throughput rate, $R$, is equal to

$$R = \sum_{j=1}^{I} R_j = \sum_{j=1}^{I} \alpha_j W_j H_{K_j}. \tag{4}$$

and the overall average throughput rate for user $k$, $R_k$, is equal to

$$R_k = \sum_{j=1}^{I} R_{k,j} = \alpha_i W_i \frac{H_{K_i}}{K_i}. \tag{5}$$

Remarks *1* and *2* capture the key properties of data cellular communication systems employing PFS [17]–[21].

*Remark 1 (Effect of Enhanced Multiuser Diversity on $R_i$):* Increasing the number of system users increases the system multiuser diversity, which increases the average cell throughput rate in data cellular communication systems employing PFS, i.e. $H_K$ increases monotonically.

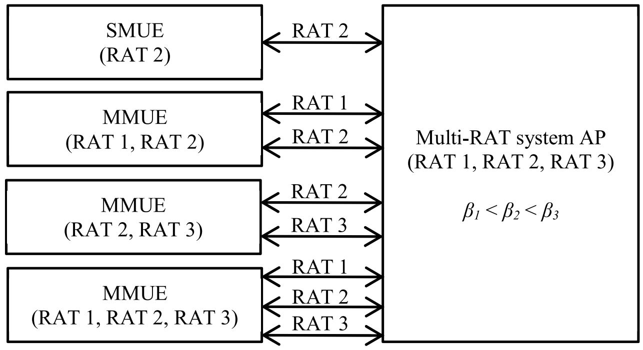


**FIGURE 6. An example of MMA in multi-RAT systems, in which the same setup adopted by Fig. 3 is assumed. Unlike SMA, MMUE utilize all RATs jointly supported by MMUE and the connecting system AP under MMA.**

*Remark 2 (Effect of Enhanced Multiuser Diversity on $R_{k,i}$):* Increasing the number of system users reduces average user access to system radiofrequency resources, thus degrading the average user throughput rate in data cellular communication systems employing PFS, i.e. $H_K/K$ decreases monotonically.

## III. ENHANCING THE PERFORMANCE OF MULTI-RAT SYSTEMS UNDER INDEPENDENT RAT OPERATION

### A. MULTIMODE ACCESS FOR MMUE

Rather than restricting MMUE to use a single RAT when connecting to system APs, MMUE simultaneously utilize all RATs jointly supported by the MMUE and the connecting AP, as Fig. 6 shows, when connecting to system APs under Multimode Access (MMA). To avoid imposing implementation requirements on the NI functionality when enabling MMA, i.e. maintain independent RAT operation, traffic request split between utilized RATs, along with the aggregation of received data streams from different RATs, is managed by the application software of MMUE [26], [28]. Let $\boldsymbol{I}_{\mathbf{k}}$ denote the set of RATs jointly supported by the connecting AP and the UE transceiver module of user $k$, with $|\boldsymbol{I}_{\mathbf{k}}| \leq I$, and let $K_i'$ denote the number of users capable of utilizing RAT $i$ at the considered cell, with $K_i < K_i' < K$. Enabling MMA thus increases the number of active users utilizing employed RATs and alters the average cell and user throughput rates, under FSA and IRA, to be equal to

$$R_i|_{MMA} = \alpha_i W_i H_{K_i'}. \tag{6}$$

$$R_{k,i}|_{MMA} = \frac{R_i|_{MMA}}{K_i'} = \alpha_i W_i \frac{H_{K_i'}}{K_i'}. \tag{7}$$

$$R|_{MMA} = \sum_{j=1}^{I} R_j|_{MMA} = \sum_{j=1}^{I} \alpha_j W_j H_{K_j'}. \tag{8}$$

$$R_k|_{MMA} = \sum_{j=1}^{I} R_{k,j}|_{MMA} = \sum_{j \in I_k} \alpha_j W_j \frac{H_{K_j'}}{K_j'}. \tag{9}$$

The gain in the average cell throughput rate of RAT $i$ due to the employment of MMA over SMA, under FSA and IRA,

$G_i|_{MMA}$, is defined as

$$G_i|_{MMA} = \frac{R_i|_{MMA}}{R_i} = \frac{H_{K_i'}}{H_{K_i}}. \tag{10}$$

Similarly, gains in the overall average user and cell throughput rates due to the employment of MMA over SMA, denoted $G_k|_{MMA}$ and $G_T|_{MMA}$, respectively, are defined as

$$G_k|_{MMA} = \frac{R_k|_{MMA}}{R_k} = \frac{\sum_{j \in I_k} \alpha_j W_j \frac{H_{K_j'}}{K_j'}}{\alpha_i W_i \frac{H_{K_i}}{K_i}}. \tag{11}$$

$$G_T|_{MMA} = \frac{R|_{MMA}}{R} = \frac{\sum_{j=1}^{I} \alpha_j W_j H_{K_j'}}{\sum_{j=1}^{I} \alpha_j W_j H_{K_j}}. \tag{12}$$

*Remark 3 (Average Cell Throughput Rate Gains Under MMA):* When compared to SMA under IRA, enabling MMA improves the average cell throughput rate of all employed RATs in multi-RAT data cellular systems employing PFS.

*Verification:* By definition, $K_i' > K_i$

$\Rightarrow$ From Remark *1*, $H_{K_i'} > H_{K_i}$

$\Rightarrow G_i|_{MMA} > 1$

$\Rightarrow R_i|_{MMA} > R_i$ ■

*Remark 4 (Overall Average Cell Throughput Rate Gains Under MMA):* When compared to SMA under IRA, enabling MMA improves the overall average cell throughput rate of multi-RAT data cellular systems employing PFS.

*Verification:* Follows from Remark *3*. ■

*Remark 5 (SMUE Performance Under MMA):* When compared to SMA under IRA, enabling MMA degrades the overall average user throughput rate of SMUE in multi-RAT data cellular systems employing PFS.

*Verification:* For SMUE, $\sum_{j \in I_k} \alpha_j W_j \frac{H_{K_j'}}{K_j'} = \alpha_i W_i \frac{H_{K_i'}}{K_i'}$.

$$\Rightarrow G_k|_{MMA} = \left(H_{K_i'}/K_i'\right) / \left(H_{K_i}/K_i\right)$$

By definition $K'_i > K_i$

$\Rightarrow$ From Remark 2, $\left(H_{K_i'}/K_i'\right) < \left(H_{K_i}/K_i\right)$

$\Rightarrow G_k|_{MMA} < 1$

$\Rightarrow R_k|_{MMA} < R_k$ ■

*Remark 6 (MMA Gains for Systems Employing SMUE):* MMA gains in multi-RAT data cellular systems employing PFS diminish as system users predominantly employ SMUE.

*Verification:* As $H_{K_i'} \to H_{K_i}$, $G_i|_{MMA} \to 1$, $G_k|_{MMA}, \to 1$ and $G_T|_{MMA} \to 1$. ■

### B. AUTONOMOUS SPECTRUM ASSIGNMENT AT SYSTEM APs

While MMA provides a UE-centric approach to enhancing the performance of multi-RAT systems that avoids imposing implementation requirements on the system NI, the trade-off of enhancing the performance of MMUE at the expense of degrading the performance of SMUE is introduced when enabling MMA in multi-RAT systems. More importantly, MMA gains are highly dependent on the system MMUE density and diminish when system users primarily employ SMUE. An alternative approach to enhancing the performance of multi-RAT systems is the autonomous assignment of spectrum at system APs. Unlike system-level FSA, Spectrum is autonomously assigned at system APs, based on AP traffic conditions, under Autonomous Spectrum Assignment (ASA). In addition to the independent assignment of spectrum at system APs, spectrum partitioning between co–deployed RATs under ASA is continuously adapted to changing system traffic conditions as Fig. 7 shows.

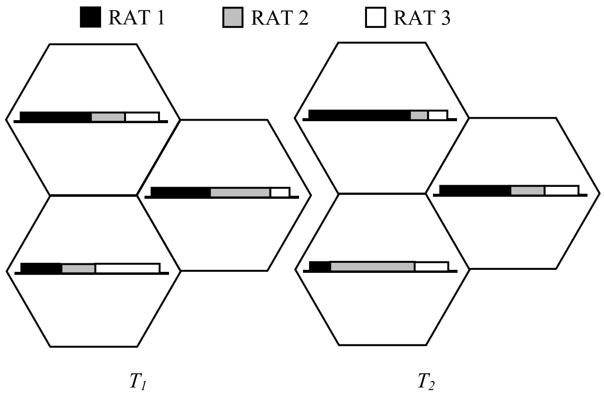


**FIGURE 7.** An example of spectrum assignment under ASA at two different times ($T_1$ and $T_2$), in which the same setup adopted by Fig. 4 is assumed. Spectrum is independently assigned at each system AP, with spectrum assignment dynamically adapted to traffic conditions.

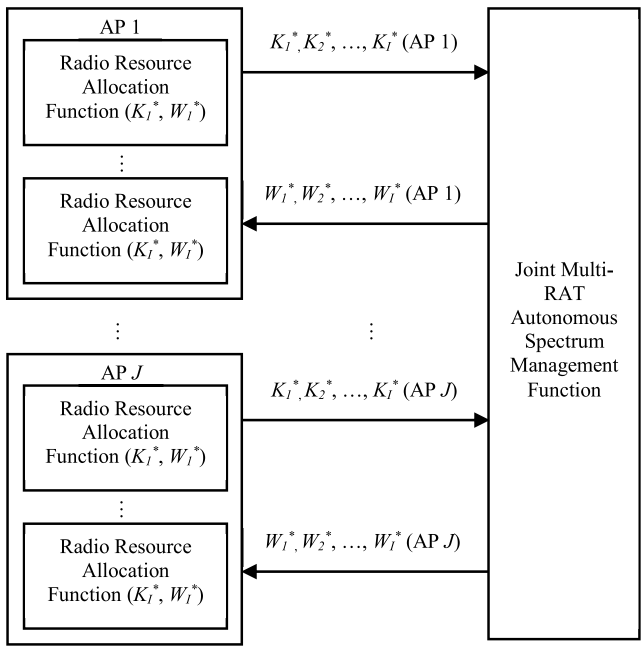


**FIGURE 8.** Illustration of the autonomous spectrum assignment function in a multi-RAT system co-deploying $I$ RATs at $J$ APs. The autonomous spectrum management function determines spectrum partitioning between co-deployed RATs at each system AP based on reported traffic demand.

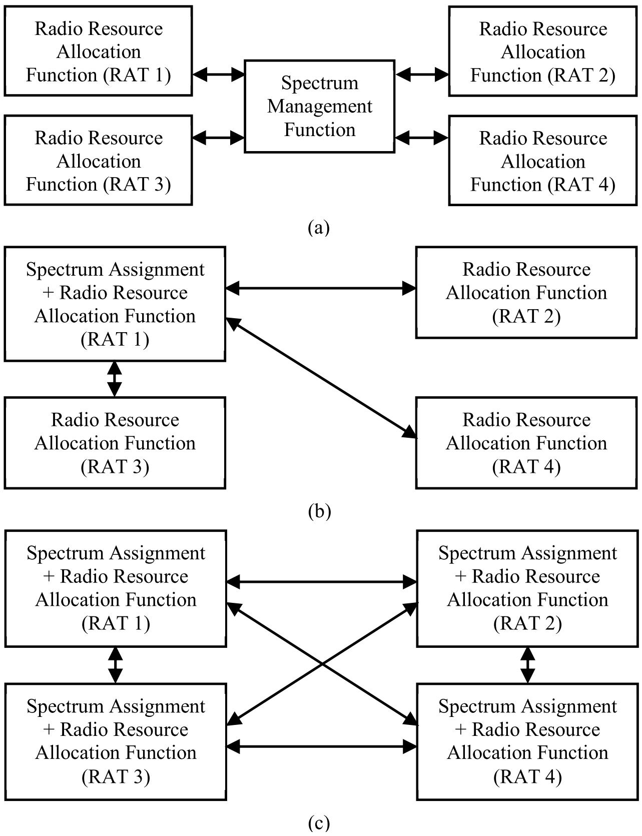


**FIGURE 9. Illustration of possible implementations of the joint multi-RAT autonomous spectrum management function in a multi-RAT system co-deploying four RATs: (a) Centralized implementation using an independent RAN element. (b) Master-Slave centralized implementation, with the master functionality being implemented in RAT 1. (c) Distributed implementation.**

Enabling ASA under IRA thus requires the introduction of a joint multi-RAT autonomous spectrum management function, illustrated in Fig. 8. Radio resource allocation functions of employed RATs report RAT traffic demand at each system AP to the autonomous spectrum management function. The autonomous spectrum management function then determines spectrum partitioning between co-deployed RATs at each system AP and returns AP spectrum assignment configurations to the radio resource allocation function of each RAT, thus ensuring that spectrum partitioning between co-deployed RATs reflects traffic conditions at all system APs [23], [27]. As Fig. 9 shows, implementation of the joint multi-RAT autonomous spectrum management function can be centralized or distributed. Matching spectrum assignment to traffic conditions at system APs thus alters the transmission bandwidth assigned to employed RATs at different system APs. In particular, the transmission bandwidth assigned to RAT $i$ at the considered system cell under ASA, $W_i^*$, is equal to

$$W_i^* = \frac{K_i}{K} W. \tag{13}$$

The average cell and user throughput rates under ASA, assuming SMA and IRA, are thus altered to

$$R_i|_{ASA} = \alpha_i W_i^* H_{K_i}. \tag{14}$$

$$R_{k,i}|_{ASA} = \frac{R_i|_{ASA}}{K_i} = \alpha_i W_i^* \frac{H_{K_i}}{K_i}. \tag{15}$$

$$R|_{ASA} = \sum_{j=1}^{I} R_j|_{ASA} = \sum_{j=1}^{I} \alpha_j W_j^* H_{K_j}. \tag{16}$$

$$R_k|_{ASA} = \sum_{j=1}^{I} R_{k,j}|_{ASA} = \alpha_i W_i^* \frac{H_{K_i}}{K_i}. \tag{17}$$

Hence $G_i|_{ASA}$, the gain in the average cell throughput rate for RAT $i$ due to the employment of ASA over FSA, assuming SMA and IRA, is equal to

$$G_i|_{ASA} = \frac{R_i|_{ASA}}{R_i} = \frac{W_i^*}{W_i} = \frac{K_i/K}{K_{T,i}/K_T}. \tag{18}$$

Similarly, $G_k|_{ASA}$, the gain in the overall average user throughput rate due to the employment of ASA over FSA, is equal to

$$G_k|_{ASA} = \frac{R_k|_{ASA}}{R_k} = \frac{W_i^*}{W_i} = \frac{K_i/K}{K_{T,i}/K_T}. \tag{19}$$

and the gain in the overall average cell throughput rate, $G_T|_{ASA}$, is equal to

$$G_T|_{ASA} = \frac{R|_{ASA}}{R} = \frac{K_T}{K} \frac{\sum_{j=1}^{I} \alpha_j K_j H_{K_j}}{\sum_{j=1}^{I} \alpha_j K_{T,j} H_{K_j}}. \tag{20}$$

*Remark 7 (Average Cell Throughput Rate Gains Under ASA):* When compared to FSA under SMA and IRA, enabling ASA in multi-RAT data cellular systems employing PFS improves the average cell throughput rate of RATs with increasing traffic demand while degrading the average cell throughput rate of RATs with declining traffic demand.

*Verification:* When $(K_i / K) > (K_{T,i} / K_T)$, $G_i|_{ASA} > 1$
$\Rightarrow R_i|_{ASA} > R_i$
Conversely, When $(K_i/K) < (K_{T,i} / K_T)$, $G_i|_{ASA} < 1$
$\Rightarrow R_i|_{ASA} < R_i$ ■

*Remark 8 (Overall Average Cell Throughput Rate Gains Under ASA):* When compared to FSA under SMA and IRA, enabling ASA in multi-RAT data cellular systems employing PFS improves the overall average cell throughput rate as the number of system users employing the highest performing RAT increases.

*Verification:* Let $v$ denote the highest performing RAT in a multi-RAT system, i.e. $\alpha_v > \alpha_i$.

$$\Rightarrow \sum_{j=1}^{I} \alpha_j \frac{K_j}{K} H_{K_j} \geq \sum_{j=1}^{I} \alpha_j \frac{K_{T,j}}{K_T} H_{K_j} \text{ as} (K_v/K) \to 1$$

$$\Rightarrow G_T|_{ASA} \geq 1 \text{ as} (K_v/K) \to 1$$

$$\Rightarrow R|_{ASA} > R \text{ as } (K_v/K) \to 1$$ ■

*Remark 9 (Average User Throughput Rate Gains Under ASA):* When compared to FSA under SMA and IRA, enabling ASA in multi-RAT data cellular systems employing PFS improves the average throughput rate of users employing RATs with increasing traffic demand while degrading the average throughput rate of users employing RATs with declining traffic demand.

*Verification:* When $(K_i / K) > (K_{T,i} / K_T)$, $G_k|_{ASA} > 1$ $\Rightarrow R_k|_{ASA} > R_k$

Conversely, When $(K_i / K) < (K_{T,i} / K_T)$, $G_k|_{ASA} < 1$ $\Rightarrow R_k|_{ASA} < R_k$ ■

In practice, the employed RATs must have comparable carrier power densities to avoid degrading user Signal to Interference plus Noise Ratio (SINR) [23], [27], [30]. Furthermore, a minimal transmission bandwidth must be continuously assigned to each of the employed RATs at all system APs to maintain user connectivity [2], [7]–[9], [23]. Let $C_i$ denote the carrier bandwidth of RAT $i$, defined as the minimal bandwidth required to connect system users utilizing RAT $i$. The maximum transmission bandwidth available for the spectrum management function at any system AP, referred to as the ASA transmission budget and denoted $W'$, is equal to

$$W' = W - \sum_{j=1}^{I} C_j. \tag{21}$$

The approximation $W' \cong W$ can be assumed when $W \gg C_i$ [7]–[9], [27]. $W'$ is partitioned into pooled spectrum blocks that can be assigned to any of the employed RATs. Maximizing pooled spectrum assignment flexibility, i.e. the number of pooled spectrum blocks, while ensuring that pooled spectrum blocks can be assigned to any of the employed RATs requires setting the pooled spectrum block bandwidth, $W_B$, to the value

$$W_B = LCM(C_1, C_2, \ldots, C_I). \tag{22}$$

in which LCM denotes the Least Common Multiple [27]. The number of pooled spectrum blocks, $B$, is thus equal to

$$B = \left\lfloor \frac{W'}{W_B} \right\rfloor. \tag{23}$$

To accommodate varying transmission frame durations of employed RATs, ASA defines a minimal pooled spectrum block assignment interval, denoted $TA_{min}$. Let $T_i$ denote the transmission frame duration of RAT $i$. $TA_{min}$ is set to the value

$$TA_{min} = LCM(T_1, T_2, \ldots, T_I). \tag{24}$$

Spectrum assignment adaptation at any system AP is achieved by altering the assignment of pooled spectrum blocks at different pooled spectrum block assignment intervals. The number of pooled blocks assigned to RAT $i$ during pooled spectrum block assignment interval $n$, $B_i(n)$, is equal to

$$B_i(n) = \left\lfloor \frac{K_i(n)}{\sum_{j=1}^{I} K_j(n)} \right\rfloor. \tag{25}$$

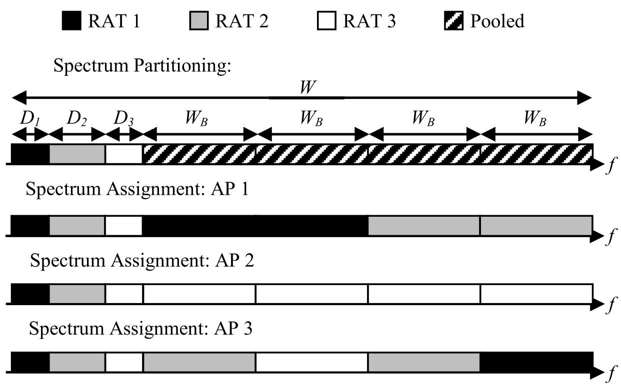


**FIGURE 10.** An example of spectrum assignment at a pooled spectrum block assignment interval under ASA in multi-RAT systems, in which the same setup adopted by Fig. 4 is assumed. Transmission bandwidths $D_1 \geq C_1$, $D_2 \geq C_2$ and $D_3 \geq C_3$ are continuously assigned to RATs 1, 2 and 3, respectively, at all system APs to maintain user connectivity. The remaining radiofrequency resources are divided into four blocks of equal bandwidth, $W_B$, and pooled spectrum blocks are autonomously assigned at each system AP based on AP traffic proportions as defined by (25). Pooled spectrum block assignment can be contiguous or non-contiguous.

in which $K_i(n)$ is the number of users utilizing RAT $i$ as the primary mode of operation during pooled spectrum block assignment interval $n$. Therefore, the total transmission bandwidth assigned to RAT $i$ during pooled spectrum block assignment interval $n$, $W_i^*(n)$, is equal to

$$W_i^*(n) = D_i + W_B B_i(n) = D_i + W_B \left\lfloor \frac{K_i(n)}{\sum_{j=1}^{I} K_j(n)} \right\rfloor. \tag{26}$$

in which $D_i \geq C_i$ is the transmission bandwidth continuously assigned to RAT $i$ to maintain user connectivity, with $W_B B + \Sigma_I D_i = W$. To avoid restricting the number of pooled spectrum blocks that can be assigned to any of the employed RATs, the NI hardware and links must support the allocation of all pooled spectrum blocks to any of the employed RATs at any pooled spectrum block assignment interval [23], [27]. An example of ASA spectrum partitioning and assignment at a pooled spectrum block assignment interval is provided in Fig. 10.

## IV. UNIFIED STRUCTURING AND RAT OPERATION IN MULTI-RAT SYSTEMS

Unification of RAN structuring in multi–RAT systems is realized by eliminating the redundant duplication of non-radio transmission RAN functions and elements such that all RAN functions, aside from radio transmission functions, are jointly performed for all employed RATs as Fig. 11 shows. All RATs thus benefit from enhancements to unified non-radio transmission RAN functions as opposed to requiring the enhancement of individual non-radio transmission RAN functions of each RAT under disjoint RAN structuring. Furthermore, the introduction of new RATs is facilitated by integrating radio transmission functions of new RATs with

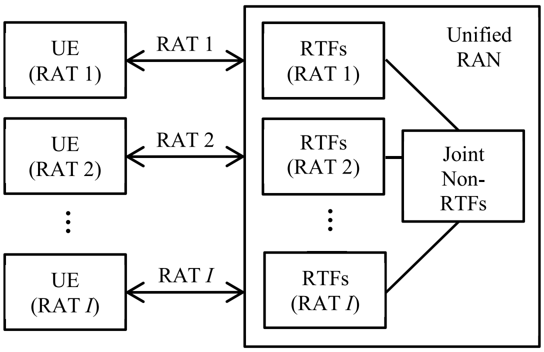


**FIGURE 11. Unified RAN structuring for a multi-RAT system employing *I* RATs.**

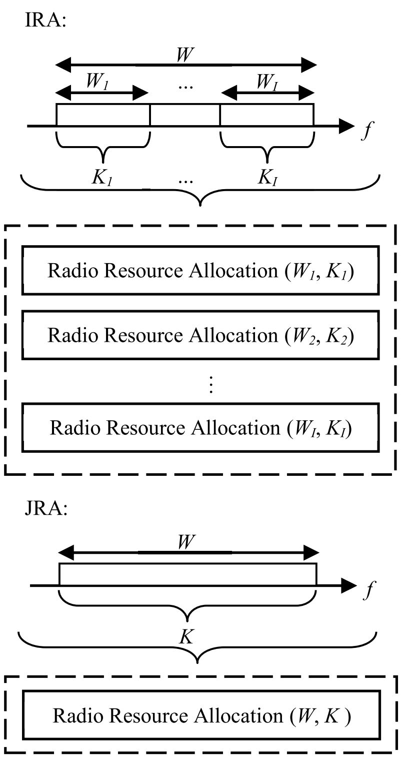


**FIGURE 12. Illustration of independent and joint resource allocation in a multi-RAT system co-deploying *I* RATs. Resource allocation is independently performed for each employed RAT under IRA using RAT-specific Spectrum. On the other hand, resource allocation is jointly performed for all system users using all system radiofrequency resources under JRA.**

the unified non-radio transmission RAN functions. Therefore, unified RAN structuring improves the scalability of multi-RAT systems while reducing deployment time, costs and operational complexity. Most importantly, the overall system performance can be enhanced through the joint implementation of non-radio transmission RAN functions, such as the joint allocation of system radiofrequency resources as Fig. 12 shows. Under Joint Resource Allocation (JRA) with ASA, system radiofrequency resources are jointly allocated for all users using the UE primary mode of operation, with the JRA function capable of assigning all system radiofrequency resources to any system user utilizing any RAT. Subsequently, the practical considerations for the realization of ASA also apply to JRA in multi-RAT systems. Unlike MMA, where system users are required to employ MMUE and utilize multiple RATs to access additional system radiofrequency resources, JRA enables the allocation of all system radiofrequency resources to any system UE using the UE primary mode of operation. Consolidating the system transmission bandwidth and user base under JRA thus alters the average cell and user throughput rates to be equal to

$$R_i|_{JRA} = \frac{K_i}{K}\alpha_i W H_K. \tag{27}$$

$$R_{k,i}|_{JRA} = \frac{R_i|_{JRA}}{K_i} = \alpha_i W \frac{H_K}{K}. \tag{28}$$

$$R|_{JRA} = \sum_{j=1}^{I} R_j|_{JRA} = \frac{W}{K}\sum_{j=1}^{I}\alpha_j K_j H_K. \tag{29}$$

$$R_k|_{JRA} = \sum_{j=1}^{I} R_{k,j}|_{JRA} = \alpha_i W \frac{H_K}{K}. \tag{30}$$

When compared to IRA under SMA, with spectrum partitioning between co-deployed RATs matching system traffic conditions, the gain in the average cell throughput of RAT $i$ under JRA, $G_i|_{JRA}$, is equal to

$$G_i|_{JRA} = \frac{R_i|_{JRA}}{R_i} = \frac{H_K}{H_{K_i}}. \tag{31}$$

and the gains in the overall average user and cell throughput rates, denoted $G_k|_{JRA}$ and $G_T|_{JRA}$, respectively, are equal to

$$G_k|_{JRA} = \frac{R_k|_{JRA}}{R_k} = \frac{H_K}{H_{K_i}}. \tag{32}$$

$$G_T|_{JRA} = \frac{R|_{JRA}}{R} = \frac{\sum_{j=1}^{I}\alpha_j K_j H_K}{\sum_{j=1}^{I}\alpha_j K_j H_{K_j}}. \tag{33}$$

*Remark 10 (Average Cell Throughput Rate Gains Under JRA):* When compared to IRA under SMA with spectrum assignment matching system traffic conditions, employing JRA improves the average cell throughput rate for all employed RATs in multi-RAT data cellular systems employing PFS.

*Verification:* By definition, $K = \sum_{j=1}^{I} K_j > K_i$
$\Rightarrow$ From Remark *1*, $H_K > H_{K_i}$
$\Rightarrow G_i|_{JRA} > 1$
$\Rightarrow R_i|_{JRA} > R_i$ ■

*Remark 11 (Overall Average Cell Throughput Rate Gains Under JRA):* When compared to IRA under SMA with spectrum assignment matching system traffic conditions, employing JRA improves the overall average cell throughput rate for all employed RATs in multi-RAT data cellular systems employing PFS.

*Verification:* Follows from Remark *10*. ■

*Remark 12 (Average User Throughput Rate Gains Under JRA):* When compared to IRA under SMA with spectrum assignment matching system traffic conditions, employing JRA improves the average throughput rate for all system users in multi-RAT data cellular systems employing PFS.

*Verification:* By definition, $K = \sum_{j=1}^{I} K_j > K_i$

$\Rightarrow$ From Remark *1*, $H_K > H_{K_i}$

$\Rightarrow G_k|_{JRA} > 1$

$\Rightarrow R_k|_{JRA} > R_k$ ■

*Remark 13 (Average Cell Throughput Rate Gains of JRA Over MMA):* When compared to IRA under MMA with spectrum assignment matching system traffic conditions, employing JRA better improves the average cell throughput rate for all employed RATs in multi-RAT data cellular systems employing PFS.

*Verification*: From (6) and (27), $\frac{R_i|_{JRA}}{R_i|_{MMA}} = \frac{H_K}{H_{K_i'}}$.

By definition, $K_i < K_i' < K$

$\Rightarrow$ From Remark *1*, $H_K > H_{K_i'}$

$\Rightarrow R_i|_{JRA} > R_i|_{MMA}$ ■

*Remark 14 (Overall Average Cell Throughput Rate Gains of JRA Over MMA):* When compared to IRA under MMA with spectrum assignment matching system traffic conditions, employing JRA better improves the overall average cell throughput rate in multi-RAT data cellular systems employing PFS.

*Verification:* Follows from Remark *13* ■

*Remark 15 (Average User Throughput Rate Gains of JRA Over MMA):* When compared to IRA under MMA with spectrum assignment matching system traffic conditions, employing JRA better improves the average user throughput rate for all employed RATs in multi-RAT data cellular systems employing PFS.

*Verification*: From (7) and (28), $\frac{R_{k,i}|_{JRA}}{R_{k,i}|_{MMA}} = \frac{K_i'}{K_i}\frac{H_K}{H_{K_i'}}$.

By definition, $K_i < K_i' < K$

$\Rightarrow$ From Remark *1*, $H_K > H_{K_i'}$

$$\Rightarrow \frac{K_i'}{K_i}\frac{H_K}{H_{K_i'}} > 1$$

$\Rightarrow R_{k,i}|_{JRA} > R_{k,i}|_{MMA}$ ■

*Remark 16 (Optimality of JRA):* When compared to different user access and spectrum assignment configurations under IRA, JRA provides the optimal performance for multi-RAT data cellular systems employing PFS.

*Verification:* Follows from Remarks *10 – 15*.

## V. SIMULATION SETUP

### A. METHODOLOGY

A system co-deploying Multi-Carrier High Speed Packet Access (HSPA) and Long Term Evolution-Advanced (LTE) [8], [9] over a contiguous 45 MHz band centered at 2 GHz is considered, with a fixed 5 MHz carrier continuously assigned to each RAT at all system APs to maintain user service continuity. System simulations, implemented using enhanced 3rd Generation Partnership Project (3GPP) system simulation models, modified to account for the effect of Inter-RAT Inter-cell interference [30], [38], are based on two nested loops: An outer shadow fading loop and an inner multipath fading loop. System users are randomly dropped, using a uniform distribution, within the system coverage area in the outer shadow fading loop, with changing user location determined through the user mobility dynamics implied by the employed channel model, and path loss, shadowing and antenna gain are calculated for each user location. Channel gain is determined in the inner multipath fading loop, along with the user SINR, and the user link performance is mapped to the corresponding system-level performance. PFS is then performed, as defined in [16], using a scheduling window size set to 1000 and a full-buffer traffic model [30]. System simulations assume an average number of 20 LTE MMUE per cell, capable of utilizing both HSPA and LTE, with various HSPA SMUE densities to emulate system traffic variations and fluctuations.

### B. SYSTEM LAYOUT AND ANTENNA MODEL

The system employs 19 wraparound sectored base stations, i.e. 57 APs, separated by an inter-site distance of 750 m and interconnected by adequate backhaul links. System UE are assumed to employ single omnidirectional antennas of 0 dBi gain. On the other hand, system APs utilize directional antennas with a radiation pattern defined by [38]

$$g(\theta_{j,k}(t)) = g_{\max} - \min(12(\theta_{j,k}(t)/\theta_{3dB})^2, A_{\max}). \quad (34)$$

in which $g(\theta_{j,k}(t))$ is the AP antenna gain, in dB, for user $k$ having an azimuth angle $\theta_{j,k}$, in degrees, with respect to AP $j$ at time $t$; $g_{\max}$ is the maximum AP antenna gain, set to 14 dBi; $\theta_{3\text{dB}}$ is the AP antenna 3dB beamwidth, set to $65^0$; $A_{\max}$ is the maximum AP antenna attenuation, set to 20 dB.

### C. PROPAGATION MODEL

The Urban macro cellular propagation model at 2 GHz, assuming an average AP antenna height of 15 m above rooftop level, is defined as [38]

$$\begin{aligned} L_P(j,k,t) = {} & 128.1 + 37.6\log_{10}(d(j,k,t)) \\ & + L_F(j,k,t) - M_F(j,k,t). \end{aligned} \quad (35)$$

in which $L_P(j, k, t)$ is the path loss, in dB, between AP $j$ and user $k$ at time $t$; $d(j, k, t)$ is the distance, in km, between AP $j$ and user $k$ at time $t$; $L_F(j, k, t)$ is the Log-normal shadowing loss, in dB, between AP $j$ and user $k$ at time $t$, modeled by a zero-mean Gaussian random variable with a standard deviation of 7 dB; $M_F(j, k, t)$ is the channel gain, in dB, due to multipath components between AP $j$ and user $k$ at time $t$, determined using the International Telecommunication Union (ITU) Vehicular A Channel Model for a user velocity of 30 km/h [39]. Therefore, the power received by user $k$ from AP $j$ at time $t$, $P_r(j, k, t)$, is equal to

$$P_r(j,k,t) = P_t(j,t) + g(\theta_{j,k}(t)) - L_P(j,k,t). \quad (36)$$

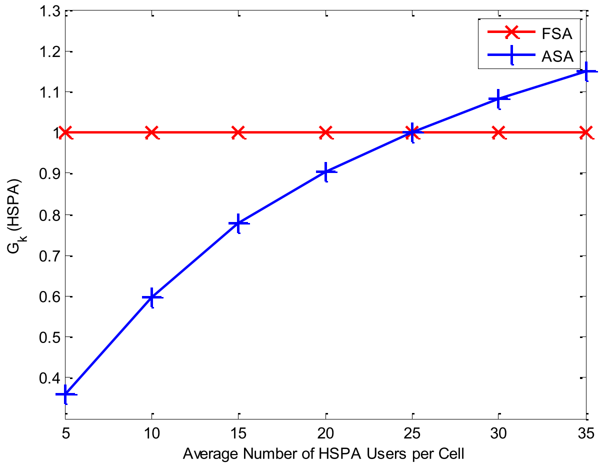


**FIGURE 13. Effect of system traffic variations on the average HSPA user throughput rate gains under ASA and FSA.**

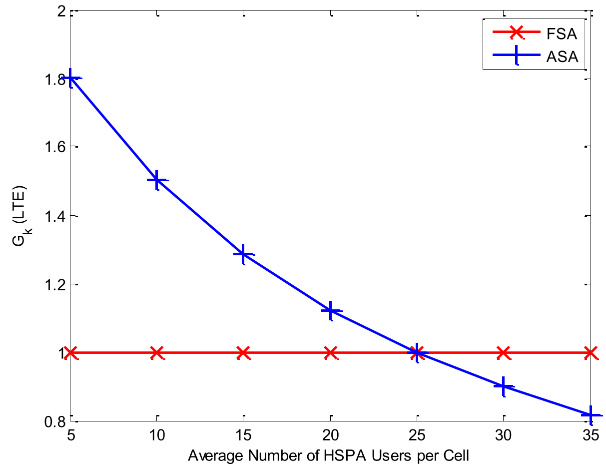


**FIGURE 14. Effect of system traffic variations on the average LTE user throughput rate gains under ASA and FSA.**

in which $P_t(j, t)$ is the transmission power of AP $j$ at time $t$, determined based on reported user channel state information.

### D. USER SINR AND THROUGHPUT RATE

SINR of user $k$ connecting to AP $j$ at time $t$, $\xi(j, k, t)$, is defined as

$$\xi(j,k,t) = (P_r(j,k,t)/(\eta_0 W + \sum_{l,l\neq j} P_r(l,k,t)))/N_F. \quad (37)$$

in which $\eta_0$ is the Additive White Gaussian Noise (AWGN) power spectral density, equal to $-174$ dBm/Hz; $W$ is the transmission bandwidth in Hz; $N_F$ is the receiver noise figure, set to 9 dB. Mapping of $\xi(j, k, t)$ to the corresponding achievable throughput rate for user $k$ connecting to AP $j$ at time $t$, $r(j, k, t)$, is facilitated by the Alpha-Shannon formula defined as [38]

$$r(j,k,t) = \begin{cases} 0, & \xi(j,k,t) < \xi_{\min} \\ \alpha W \log_2(1+\xi(j,k,t)), & \xi_{\min} \leq \xi(j,k,t) < \xi_{\max} \\ \beta W, & \xi(j,k,t) \geq \xi_{\max}. \end{cases} \quad (38)$$

in which $\alpha$ is the transmission attenuation factor for the employed RAT, set to 0.75 for LTE and 0.6 for HSPA; $\beta$ is the maximal spectral efficiency of the employed RAT, set to 4.4 bps/Hz for LTE and 4 bps/Hz for HSPA; $\xi_{\min}$ is the SINR value required to utilize the lowest performing Modulation and Coding Scheme (MCS) supported by the employed RAT, set to $-10$ dB; $\xi_{\max}$ is the SINR value required to utilize the highest performing MCS supported by the employed RAT, set to 17 dB for LTE and 20 dB for HSPA.

## VI. PERFORMANCE EVALUATION

### A. AUTONOMOUS SPECTRUM ASSIGNMENT

When comparing ASA with system-level FSA, system-level FSA is assumed to be based on an average number of 25 HSPA UE per cell, i.e. the transmission bandwidth for HSPA is assumed to be 25 MHz while the transmission bandwidth for LTE, under system-level FSA, is set 20 MHz. System HSPA and LTE cell and user performance, for various HSPA UE densities, is illustrated in Figs. 13–16. ASA assigns the majority of pooled system radiofrequency resources to LTE at low HSPA user densities, leading to substantial gains in the average LTE cell and user throughput rates at the expense of significantly degrading the HSPA average cell and user throughput rates. As the HSPA user density increases, the amount of pooled system radiofrequency resources assigned to HSPA under ASA, and thus the average HSPA cell and user performance, also increases. Conversely, the amount of pooled system radiofrequency resources assigned to LTE under ASA decreases as the HSPA user density increases, thus reducing the LTE average cell and user throughput gains. The average cell and user performance for both HSPA and LTE under ASA matches the performance under system-level FSA when the system user densities match the user densities assumed by system-level FSA. The overall average

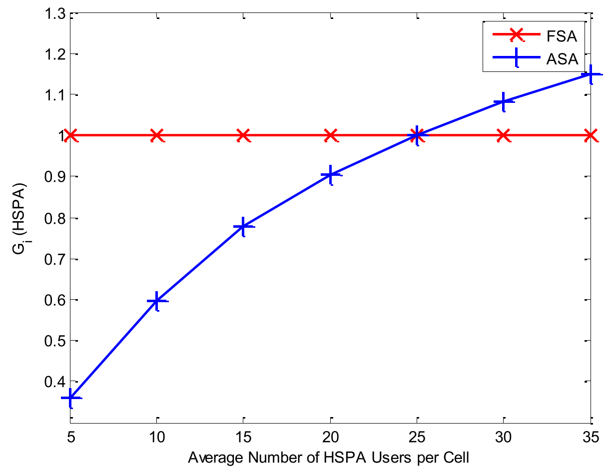


**FIGURE 15. Effect of system traffic variations on the average HSPA cell throughput rate gains under ASA and FSA.**

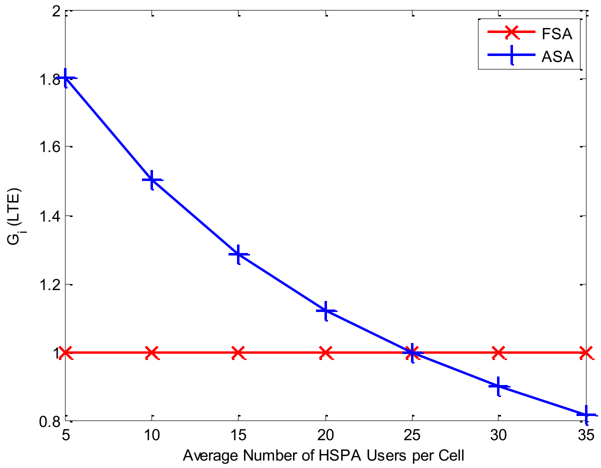


FIGURE 16. Effect of system traffic variations on the average LTE cell throughput rate gains under ASA and FSA.

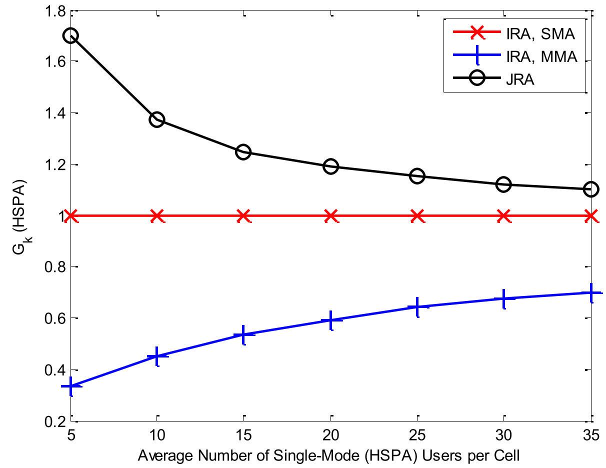


FIGURE 18. Effect of system traffic variations on the average HSPA user throughput rate gains under JRA, IRA with MMA and IRA with SMA.

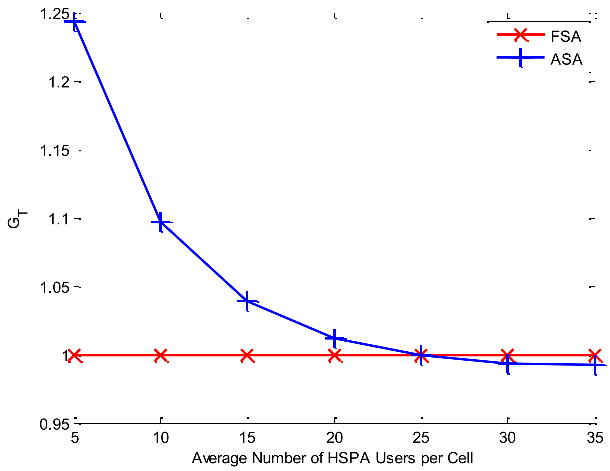


FIGURE 17. Effect of system traffic variations on the overall average cell throughput rate gains under ASA and FSA.

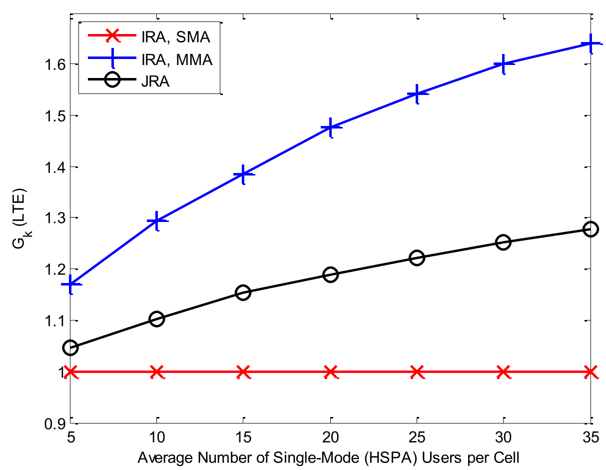


FIGURE 19. Effect of system traffic variations on the average LTE user throughput rate gains under JRA, IRA with MMA and IRA with SMA.

cell throughput performance under various HSPA user densities is shown in Fig. 17. Matching spectrum assignment to system traffic conditions substantially improves the overall average cell throughput under ASA when the system user density is primarily LTE due to the higher spectral efficiency of LTE. On the other hand, the allocation of additional system radiofrequency resources to the lower performing RAT, HSPA, results in the gradual decay in the overall system performance gains under ASA as the HSPA user density increases. Similar to the average cell and user throughput performance of HSPA and LTE, the overall system performance under ASA matches the system performance under system-level FSA when the system user densities match the user densities assumed by FSA.

### B. MULTIMODE ACCESS AND JOINT RESOURCE ALLOCATION

JRA is compared to IRA with MMA along with IRA with SMA, with spectrum partitioning between co-deployed RATs under IRA assumed to match UE densities based on the UE primary mode of operation. Figs. 18 and 19 show the average HSPA and LTE user performance for various HSPA SMUE user densities. Enabling MMA under IRA provides LTE MMUE with access to additional system radiofrequency resources and introduces the tradeoff of improving the performance of MMUE at the expense of degrading the performance of SMUE. The lower the SMUE density, the higher access MMUE have to HSPA radiofrequency resources. Therefore, degradation in the performance of SMUE under MMA is maximized at the lowest SMUE density. Increasing SMUE density reduces MMUE access to HSPA radiofrequency resources under MMA, thus reducing the degradation in the performance of HSPA SMUE. On the other hand, increasing the SMUE density also increases the amount of system radiofrequency resources assigned to HSPA, leading to higher MMUE performance gains under MMA as the SMUE density increases. The imbalance in user access to system radiofrequency resources under MMA also results in

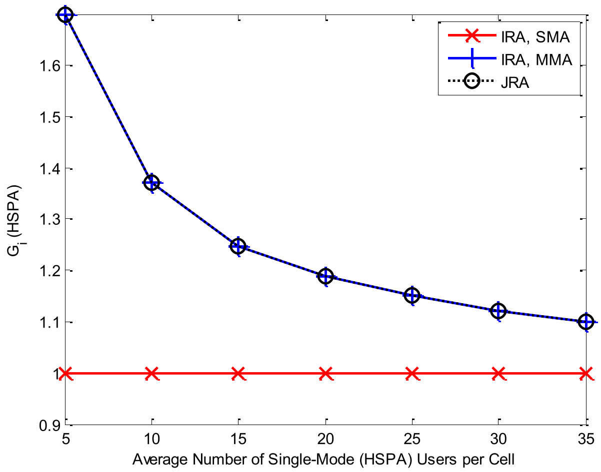


**FIGURE 20.** Effect of system traffic variations on the average HSPA cell throughput rate gains under JRA, IRA with MMA and IRA with SMA.

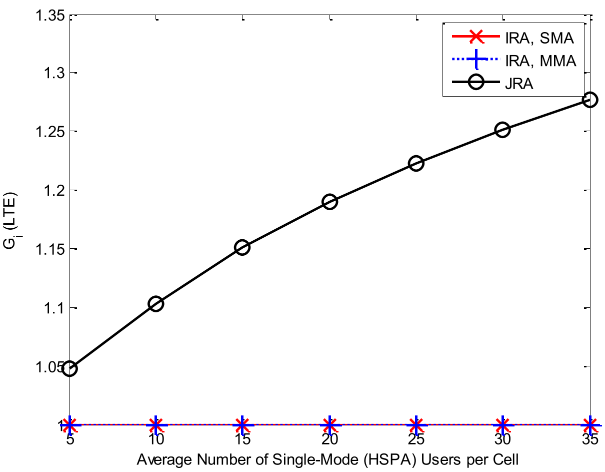


**FIGURE 21.** Effect of system traffic variations on the average LTE cell throughput rate gains under JRA, IRA with MMA and IRA with SMA.

higher performance gains for LTE MMUE under MMA with IRA when compared to JRA. Unlike MMA, where access to additional system radiofrequency resources is dependent on UE capabilities, JRA enables the allocation of all system radiofrequency resources to any system UE regardless of UE capabilities. Therefore, JRA improves the performance of both HSPA SMUE and LTE MMUE as opposed to improving the performance of LTE MMUE only under MMA. As JRA performance gains stem from enhancing system multiuser diversity, the maximal performance gains under JRA are attained when providing the highest relative increase in system multiuser diversity. Increasing the HSPA UE density decreases the relative increase in system multiuser diversity for HSPA UE under JRA. Hence the HSPA user performance gains under JRA decreases as the HSPA UE density increases. Conversely, the relative increase in system multiuser diversity for LTE UE increases with the increase in the HSPA UE density, thus increasing the LTE user performance gains under JRA as the HSPA UE density increases. The average HSPA and LTE cell throughput performance under various

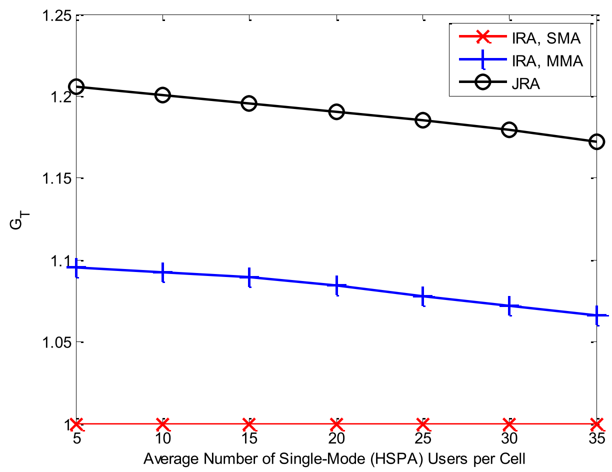


**FIGURE 22.** Effect of system traffic variations on the average LTE cell throughput rate gains under JRA, IRA with MMA and IRA with SMA.

HSPA UE densities is shown in Figs. 20 and 21. As the cell throughput performance gains under JRA are also attributed to enhanced system multiuser diversity, the average cell throughput performance gains under JRA match the user performance gains for both HSPA and LTE. Increasing the HSPA UE density thus decreases the cell throughput rate gains while increasing the LTE cell throughput rate gains under JRA. With all system users capable of utilizing HSPA, enabling MMA under IRA provides the same relative increase in HSPA multiuser diversity as JRA, leading to a similar increase in the average HSPA cell throughput under both IRA with MMA and JRA. On the other hand, as HSPA users employ SMUE, LTE multiuser diversity is unaffected by enabling MMA and, subsequently, the LTE cell throughput is also unaffected by enabling MMA. Fig. 22 shows the overall average cell throughput performance under various HSPA SMUE densities. Enhancing the average cell throughput for both HSPA and LTE under JRA, as opposed to enhancing the average cell throughput of HSPA only under IRA with MMA, allows JRA to achieve significantly higher overall average cell throughput gains when compared to IRA with MMA. Finally, it should be noted that employed traffic and user distribution models constrain the performance gains of ASA, MMA and JRA. As the system performance gains scale with multiuser diversity, higher performance gains are achieved under traffic and user distribution models with higher degrees of multiuser diversity.

## VII. CONCLUSIONS

The continuous introduction and employment of new RATs in cellular communication systems requires rethinking the structuring and operation of multi-RAT systems; as the disjoint structuring of multi–RAT systems results in limited system scalability, large system deployment costs and high operational complexity. Additionally, the employment of SMA, system-level FSA and IRA renders multi-RAT systems incapable of adapting to traffic variations and fluctuations, leading to unbalanced system loading and the suboptimal utilization of system radiofrequency resources and UE hardware capabilities. MMA provides MMUE with access to additional system radiofrequency resources and improves the overall performance of multi-RAT systems, without altering the NI functionality, by enhancing the system multiuser diversity. However, gains from employing MMA are highly dependent on the system UE capabilities and diminish when system users primarily employ SMUE. Furthermore, the tradeoff of improving the performance of MMUE at the expense of degrading the performance of SMUE is introduced when enabling MMA. The feasibility of MMA in a multi-RAT system is thus dependent on the density and performance requirements of SMUE and MMUE. To circumvent the shortcomings of system-level spectrum assignment without requiring system users to employ MMUE, ASA matches spectrum partitioning between co–deployed RATs to system traffic conditions at all system APs. Therefore, ASA gains are also dependent on the system traffic conditions, with ASA improving the performance of RATs experiencing increasing traffic demand at the expense of degrading the performance of RATs with declining traffic demand. In addition, the overall system performance under ASA is maximized when system users primarily utilize the highest performing RAT; as the majority of system radiofrequency resources would be assigned to the highest performing RAT under ASA. However, enabling the realization of ASA in a multi-RAT system imposes multiple restrictions on the transmission characteristics of employed RATs to maintain reliable system operation and maximize the spectrum assignment flexibility under ASA. The independent operation of employed RATs also entails significant modifications to the NI functionality for implementing ASA in multi-RAT systems. Nevertheless, the ability to independently assign system radiofrequency resources at system APs removes the main barrier that stands in the way of unifying radio resource allocation in multi-RAT systems. By enabling the utilization of system radiofrequency resources by all system users regardless of UE capabilities, JRA maximizes multiuser diversity and attains optimal performance for multi-RAT systems when compared to ASA and MMA under IRA. To maintain reliable system operation, the restrictions on the transmission characteristics of employed RATs under ASA are also applicable when unifying radio resource allocation in multi-RAT systems. Furthermore, the realization of JRA in multi-RAT systems requires the unification of non-radio transmission RAN functions, thus entailing the largest implementation requirements when compared to MMA and ASA. Nevertheless, when compared to disjoint multi-RAT structuring, unified RAN structuring substantially reduces system deployment costs and operational complexity while providing multi-RAT systems with the scalability to seamlessly integrate new RATs and functions in an efficient and timely manner.

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